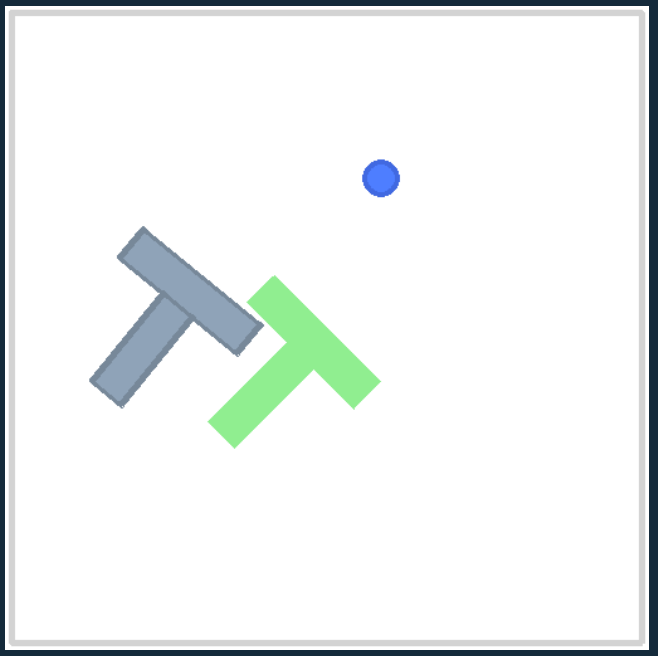
Language over demo breadcrumbs

Frozen LeWorldModel on Push-T

Words name the goal.
Demonstrations supply the steps.

A language interface for a planner
that is reliable over short hops.



Push-T · logged start state

Alexander Bashirov · Didar Seyidov · Faith Ogunwoye

Skoltech

Four things this project adds to a frozen world model.

LeWM and its planner are used unchanged. Everything new sits around them.

01

Words name the goal

A sentence becomes a goal region (on target, turn by θ, push by d) instead of one goal image. A small local LLM parses it: 101/120 unseen sentences.

02

Routes across many demos

A graph over 427,396 demo states links different demos where they pass close by, so a route exists from starts no demo began at.

03

Short hops reach far goals

The planner only ever chases a demo frame about 25 steps away. Goals 100 steps out: 10/50 → 29/50.

04

Words straight into the latent

A text-conditioned head predicts the next subgoal latent, with no graph at test time. First signal: 18/111 vs 10/111 with shuffled words.

Prior work (Hi-LeWM) used later frames of the same demo as subgoals. Every claim here was frozen in git before its scored run.

A short hop works. A distant goal is harder.

LeWM stays frozen: it encodes an image into 192 values and predicts latents under actions.

SHORT DEMO GOAL

47/50

25 steps ahead · budget 50

5090 setup check

LONG LIVE GOAL

0/20

Random live goals · budget 50

Earlier 4090 run

IN-PLACE TURN

0/20

Rendered +90° or +180° goal

Budget 50 · each angle

Course question: can a sentence name a goal this existing planner can actually follow?

These are different goal settings, not a controlled horizon sweep. Earlier 4090 short-goal baseline: 45/50.

The angle is encoded. Control is still the obstacle.

Reading a state from a latent does not guarantee the planner can reach that state.

1 Read the angle

Held-out circular error

180° frames: 12.08°
Near 0°: 14.58°

2 Add a fixed Δz

One constant latent step per command fails to fit.

Residual ratio: 0.991

3 Plan the turn

Angle-cost +90°: 2/20
Strict in-place gate: 0/20

Three rendered stages: 1/20

Decision: keep LeWM frozen and change the subgoal interface.

Make a long task a sequence of nearby goals.

Words choose a goal region. A graph of expert demonstrations supplies the intermediate frames.

1 SENTENCE → GOAL REGION

“Turn the T clockwise.”

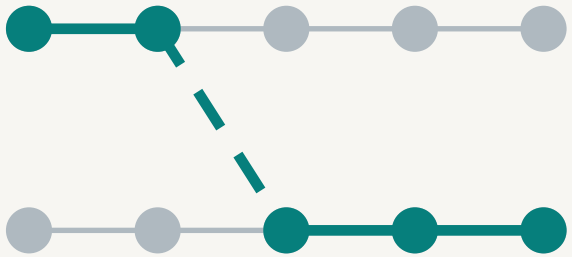
Qwen parser

→ turn +90°

Region: angle + drift tolerance

2 FIND A ROUTE

Demo A



Switch demos near matching states.

Graph drawing is schematic.

3 PLAN · ACT · REPEAT

Next demo frame

Encode frame → frozen CEM

Execute ≈25 steps

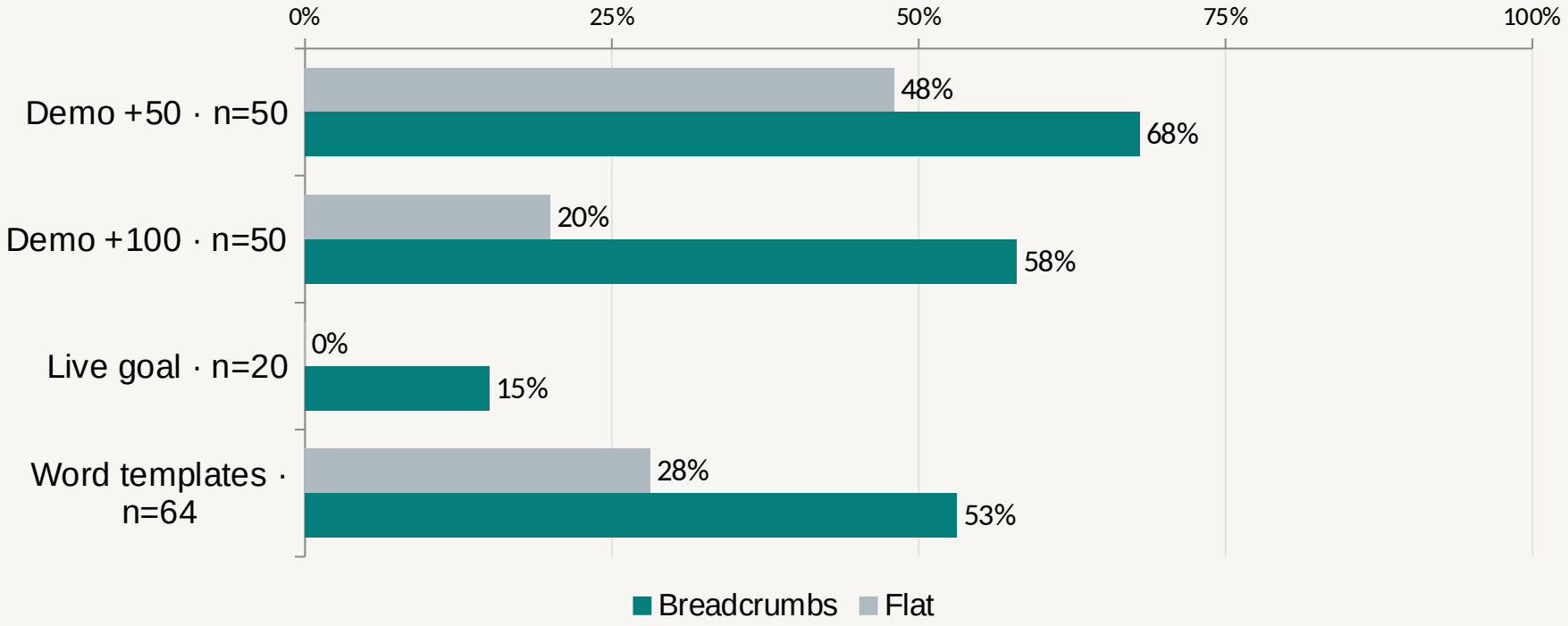
Read live state again

427,396 nodes · one node every 5 demo steps · train split only

Retrieval uses simulator state; LeWM is not trained with language.

The biggest gain appears on the farther demo goals.

Same planner and budget within each row. Only the intermediate subgoals change.



+38 pp

at offset 100

PAIRED +100 TASKS

23 gained

4 lost

19 net / 50

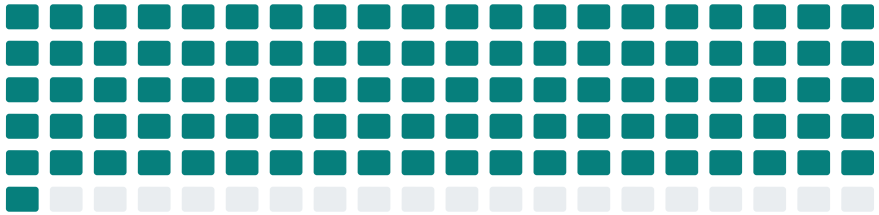
Budget 300. Rows have different frozen success criteria; compare methods within each row.

The language interface works; execution is the larger gap.

Qwen2.5-1.5B-Instruct · local, greedy · 120 held-out sentences

101/120

sentences parsed correctly · 84.2%



Keyword baseline: 31/120 · 25.8%

END-TO-END · SAME 111 ELIGIBLE TASKS

Gold command **57/111**

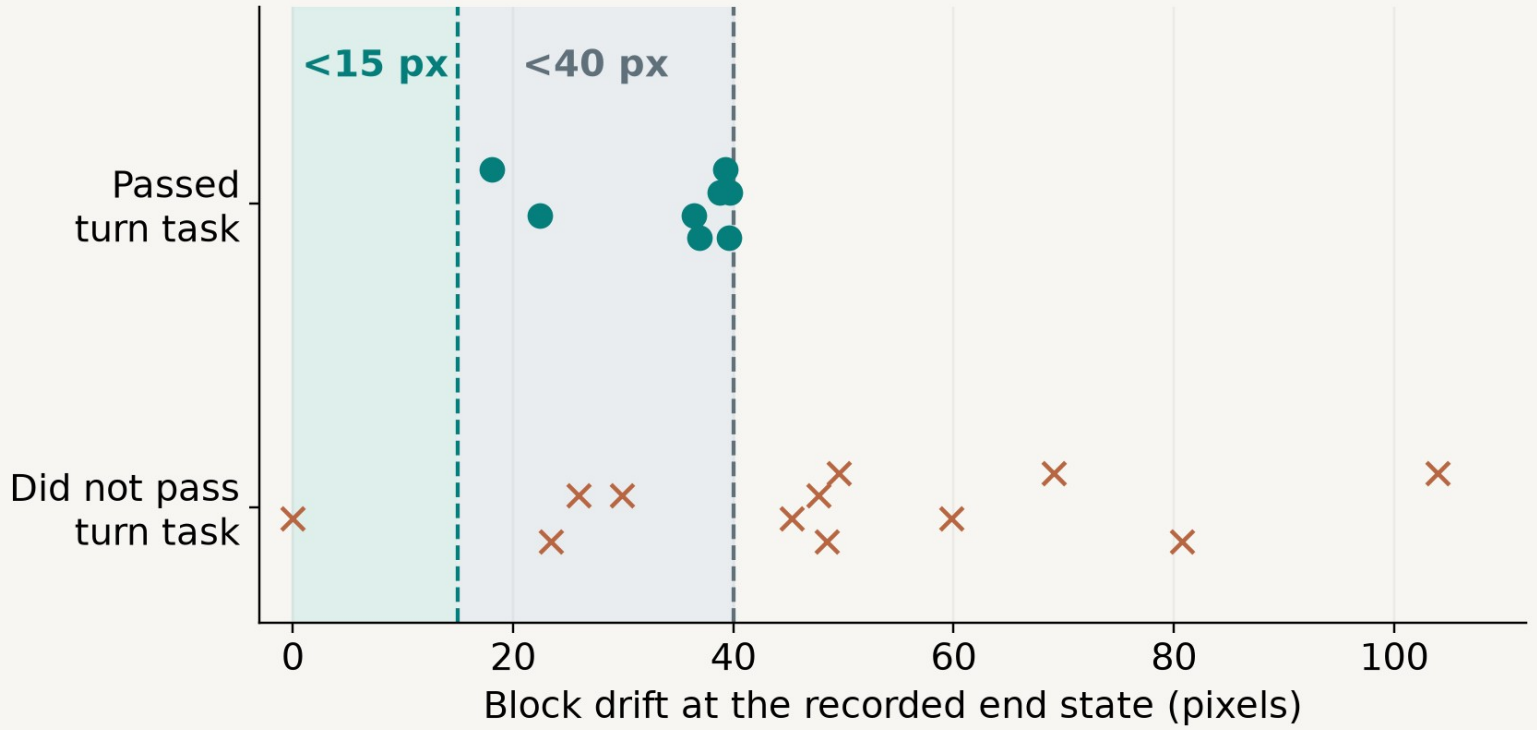
Parsed command **51/111**

Parsing costs 6 net successes (−5.4 pp).

Nine pushes excluded: gold destination outside arena. Tasks reuse 10 reset seeds; budget 300.

The T turns, but it slides while turning.

A looser task success must not be presented as an in-place rotation.



BREADCRUMBS

8/20

turn tasks reached with <40 px drift

0/20

pass the stricter <15 px in-place gate

Flat: 4/20 turns, 1/20 strict. In-place turns exist in demos: 8,040 nodes across 1,691 episodes.

Lower latent error did not deliver better control.

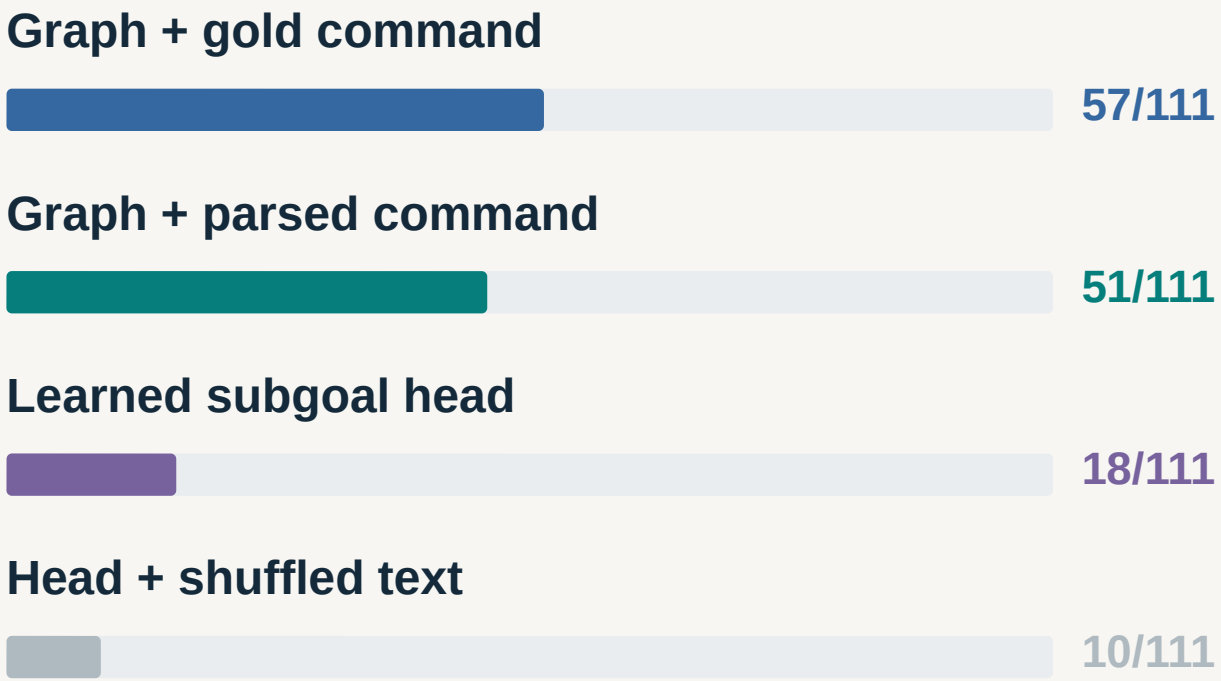
A small head predicts the next latent from start latent, current latent and a frozen text embedding.

Text → subgoal head

At test time:
no graph · no command parser

0.67×

validation MSE vs copying the current latent



Same eligible sentences and starts · budget 300

Words can name routes the frozen planner can follow.

01 **Nearby demo frames make distant goals easier.**

Offset 100: 10/50 → 29/50 with breadcrumbs.

02 **Language can select the route.**

101/120 parsed correctly; 57 → 51 successes after parsing.

03 **Reachability is still the limiting factor.**

Strict in-place turns: 0/20. Learned head: 18/111.

Frozen LeWM throughout · descriptive results on the saved task sets · no new runs for this deck

A frozen world model can follow words, one reachable hop at a time.

WHAT WE SHOWED

- Demo breadcrumbs lift success by 20–38 points on demo goals.
- Free sentences work end to end: 51/111 (57 with perfect parsing).
- Words can enter the latent directly: a first signal, mainly for “put it on the target”.

WHAT COMES NEXT

- Strict in-place turns (0/20 today)
- Random live goals (3/20 today)
- Latent head: more data, relative commands
- The lab UR3 arm with teleoperated demos

Thank you. Questions welcome.

Where did the extra successes come from?

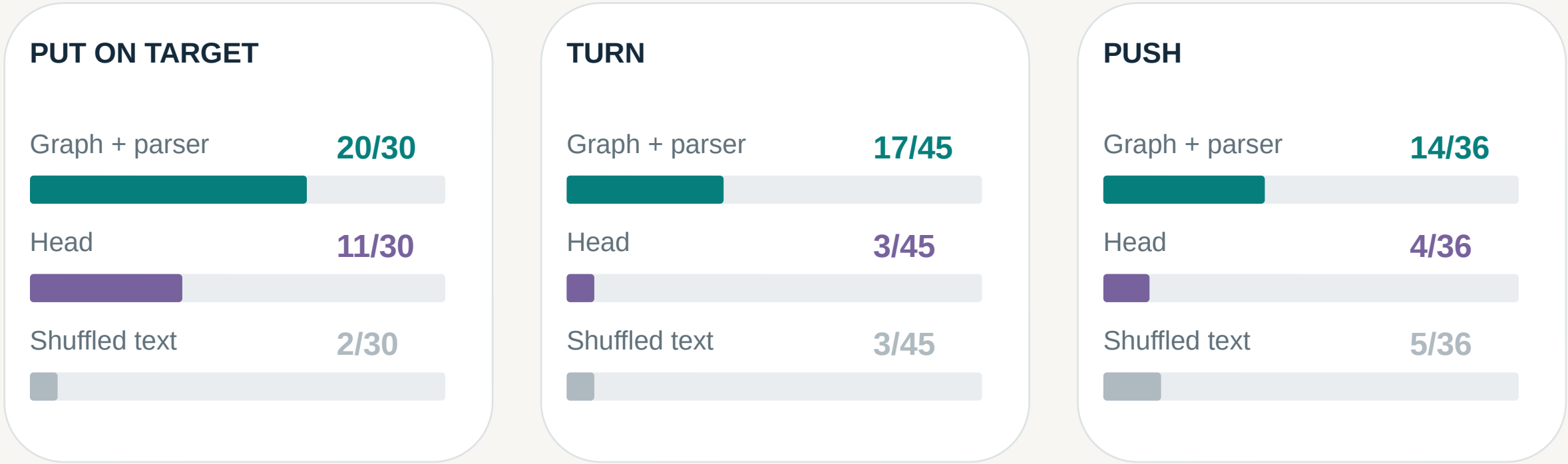
Matched tasks let us count gains and losses directly. These are descriptive counts.

TASK	BOTH SUCCEED	FLAT ONLY	CRUMBS ONLY	NEITHER
Demo +50 · n=50	18	6	**16**	10
Demo +100 · n=50	6	4	**23**	17
Live goals · n=20	0	0	**3**	17
Templates · n=64	10	8	**24**	22

Offset 100: 23 gained, 4 lost → 19 net extra successes out of 50 (+38 percentage points).

The head’s text effect is mainly on target requests.

Same 111 eligible tasks; target n=30, turns n=45, pushes n=36.



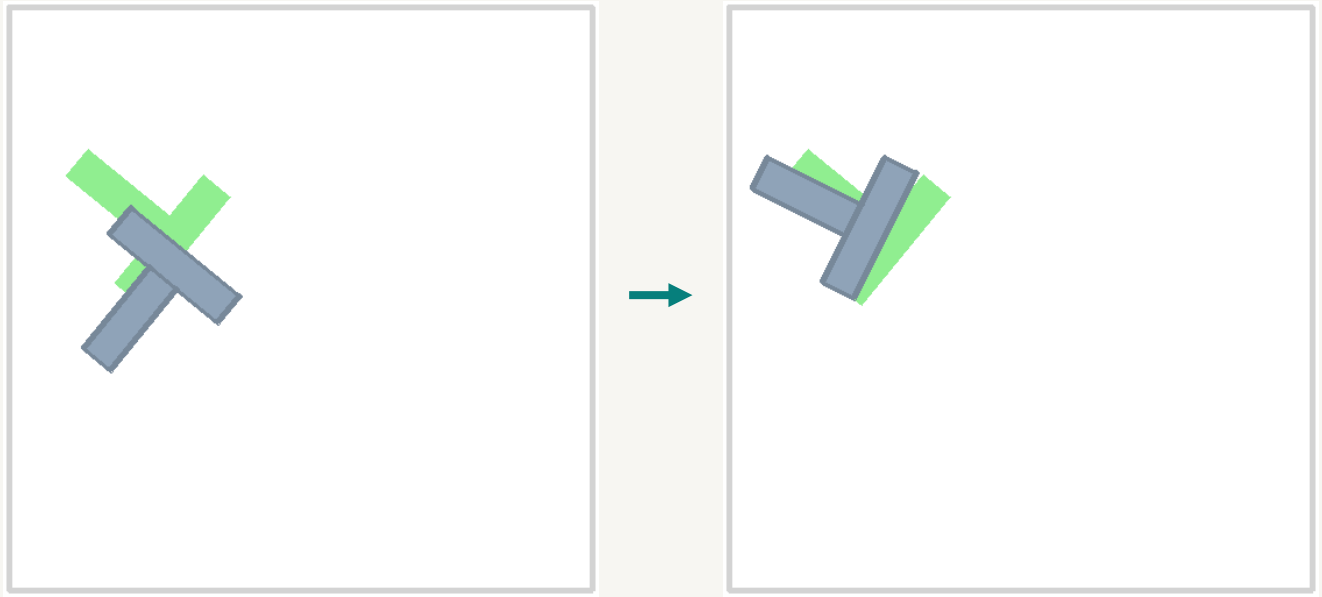
These are successes on the saved command sets, not evidence of general language understanding.

A successful quarter-turn can miss the in-place gate.

Seed 1 · clockwise turn template · breadcrumbs · requested change: +90°

START

RECORDED END



77.2°

actual clockwise turn

22.5 px

block drift

passes <40 px, fails <15 px

Reconstructed from logged block poses; pusher omitted. Green T = requested goal. No trajectory was saved.

The graph covers demo goals, not random live goals.

Training split only. One node every 5 demo steps; nearby states from different demos are linked.

427,396
demo states (nodes)

410,579
“next frame” links

2,852,207
links between demos

DEMO GOALS, +50 STEPS

50/50

have a route · median 0.5 demo switches
start lies 9 state units from a demo state

WORD COMMANDS

56/64

start/command pairs have a route
every command has matching demo states

RANDOM LIVE GOALS

13/20

have a route, but start and goal lie
37 and 39 state units from any demo state

In-place turns do occur in the demos: 8,040 nodes in 1,691 of 16,817 episodes. State units: block/pusher px, angle at 60 per radian.

Each claim and kill line was committed before its scored run.

Verdict: holds = claim met; between = claim missed, kill line not reached. Budget 300 on all scored runs.

Check	Claim (frozen first)	Kill line	Result	Verdict
0 · Setup reproduces the 4090	≈ 45/50	≤ 40/50	47/50	holds
1a · Routes exist, demo goals	≥ 35/50	< 35/50	50/50	holds
2b · Breadcrumbs, +50 steps	flat + 10 pp	≤ flat + 2 pp	68% vs 48%	holds
2c · Breadcrumbs, +100 steps	gap grows	crumbs ≤ flat	58% vs 20%	holds
2d · Random live goals	≥ 4/20	≤ flat + 1	3/20 vs 0/20	between
3b · Word templates	flat + 10 pp	≤ flat + 1 ep	34 vs 18 of 64	holds
3c · Turns (40 px drift allowed)	≥ 5/20	≤ 2/20	8/20	holds
L1 · Sentence parsing	≥ 90%	≤ 75%	84.2%	between
L2 · Parsed vs gold, end to end	≥ gold − 5 pp	≤ gold − 15 pp	−5.4 pp	between
S1 · Head, offline error	≤ 0.7× copy	≥ 0.9× copy	0.67×	holds
S2 · Head, end to end	≥ 34/111	≤ 11/111	18/111	between
S3 · Words matter for the head	shuffled + 10	≤ shuffled + 3	+8	between

Sources

[1] LeCun, Y. (2022). A Path Towards Autonomous Machine Intelligence. OpenReview, v0.9.2.

[2] Assran, M., et al. (2023). Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture (I-JEPA). CVPR.

[3] Maes, L., Le Lidec, Q., Scieur, D., LeCun, Y., Balestriero, R. (2026). LeWorldModel: Stable End-to-End Joint-Embedding Predictive Architecture from Pixels. arXiv:2603.19312.

[4] Caselli, Massafra, Punzo, Lo Sardo, Pantelidis, Bhethanabhotla (2026). Hi-LeWM. arXiv:2607.12547.

[5] Florence, P., et al. (2021). Implicit Behavioral Cloning. CoRL. (Push-T task)

[6] Rubinstein, R. Y. (1999). The Cross-Entropy Method for Combinatorial and Continuous Optimization. Methodology and Computing in Applied Probability.

[7] Yang, A., et al. (2024). Qwen2 Technical Report. arXiv:2407.10671. Model used: Qwen2.5-1.5B-Instruct (Qwen Team, 2024).

[8] Myers, V., et al. (2023). Goal Representations for Instruction Following (GRIF). CoRL. arXiv:2307.00117.

[9] Chen, D., et al. (2025). VL-JEPA: Joint Embedding Predictive Architecture for Vision-Language. arXiv:2512.10942.

Code, frozen claims and all result files: github.com/velosipedistufa/JEPA · branch 5090-run-demo-breadcrumbs